1.45 Proof

Worked Solutions

1 a)

$A \Leftarrow B$

1 b)

$A \Leftarrow B$

1 c)

$A \Leftrightarrow B$

1 d)

$A \Rightarrow B$

2 a)

$A \Rightarrow B$

2 b)

$A \Leftarrow B$

2 c)

$A \Rightarrow B$

2 d)

$A \Leftrightarrow B$

3 a)

True

3 b)

True

3 c)

False, not true for $x = 0$

3 d)

True

3 e)

False. It is always odd.

3 f)

False as x could be another value such as 300 for example

4)

At the point $\frac{(10-10)(10+10)}{10(10-10)}$ there is a zero in the denominator

5 a)

2 is a prime number and its even.

5 b)

$3^2 = 9$ which is odd.

5 c)

3 is a multiple of 3 but not a multiple of 6

5 d)

-1 does not have a real square root.

5 e)

$4 = 2^2$ ends in an even number.

6 a)

For $x = -2$, $x^4 = (-2)^4 = 16$

$16 > -2$ and $-2 \not> 1$

$\therefore$ not true for $x = -2$

6 b)

When $n = 13$, $13^2 - 13 + 13 = 13^2 = 169$ which is not a prime number.

6 c)

When $n = 16$,

$16^2 + 16 + 17 = 16(16 + 1) + 17 = 17^2 = 289$ which is not a prime number.

7 a)

Combinations to form a 2 digit number:

+	0	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9	10
2	2	3	4	5	6	7	8	9	10	11
3	3	4	5	6	7	8	9	10	11	12
4	4	5	6	7	8	9	10	11	12	13
5	5	6	7	8	9	10	11	12	13	14
6	6	7	8	9	10	11	12	13	14	15
7	7	8	9	10	11	12	13	14	15	16
8	8	9	10	11	12	13	14	15	16	17
9	9	10	11	12	13	14	15	16	17	18

Therefore, the sum of the digits of any 2-digit number is less than or equal to 18

7 b)

Combinations to form a 2 digit number, which are divisible by 3:

+	0	1	2	3	4	5	6	7	8	9
1			3			6			9	
2		3			6			9		
3	3			6			9			12
4			6			9			12	
5		6			9			12		
6	6			9			12			15
7			9			12			15	
8		9			12			15		
9	9			12			15			18

Therefore if a 2-digit number is divisible by 3, then the sum of its digits is divisible by 3

7 c)

$\sqrt{89} = 9.43\ldots$ so check prime numbers $< 9.43\ldots$

$$89 \div 7 = 12.71\ldots$$

$$89 \div 5 = 17.8$$

$$89 \div 3 = 29.66\ldots$$

$$89 \div 2 = 4.5$$

None of these are integers, so 89 is a prime number.

8)

Consider the end digits of every number

+	0	1	2	3	4	5	6	7	8	9
$\wedge 2$	0	1	4	9	6	5	6	9	4	1

None of these have 3 as the final digit, so we have proved the original statement that no square number ends in a 3

9 a)

Two consecutive numbers: $2k$ and $2k + 1$

$$(2k) + (2k + 1) = 4k + 1$$

$\therefore$ the sum of 2 consecutive numbers is always odd

9 b)

Two consecutive numbers: $2k + 1$ and $2k + 3$

$$(2k + 1) + (2k + 3) = 4k + 4 = 4(k + 1)$$

$\therefore$ the sum of any two consecutive odd numbers is a multiple of 4

9 c)

Odd number: $2k+1$

$$(2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$$

$\therefore$ The square of an odd number is always odd.

9 d)

Two consecutive odd numbers: $2k+1$ and $2k+3$

$$\begin{aligned}&(2k+1)^2 + (2k+3)^2\\ &= 4k^2 + 4k + 1 + 4k^2 + 12k + 9\\ &= 8k^2 + 16k + 10\\ &= 2(4k^2 + 8k + 5)\end{aligned}$$

$\therefore$ The sum of the squares of 2 consecutive odd numbers is always even.

10 a)

Three consecutive positive integers:

$$k, k+1, k+2$$

At least one of these is even so divisible by 2

One of these must be a multiple of 3

As the expression is a multiple of both 2 and 3, it must be a multiple of 6

10 b)

Four consecutive positive integers:

$$k, k+1, k+2, k+3$$

At least two of these are even:

One must be divisible by 2 the other by 4

One of these must be a multiple of 3

As the expression is a multiple of both 2, 3 and 4, it must be a multiple of 24

11)

Since n is even: $n = 2k$

$$\begin{aligned}6n^2 + 10n = 6(2k)^2 + 10(2k) &= 24k^2 + 20k\\ &= 4(6k^2 + 5k)\end{aligned}$$

$\therefore 6n^2 + 10n$ has a factor of 4

12)

$$\begin{aligned}n^4 - n^2 = n^2(n^2 - 1) &= n^2(n-1)(n+1)\\ &= n \times [(n-1) \times n \times (n+1)]\end{aligned}$$

So it is $n\times$ the product of three consecutive integers.

Three consecutive positive integers:

$$n-1, n, n+1$$

At least one of these is even so divisible by 2

One of these must be a multiple of 3

As the expression is a multiple of both 2 and 3, it must be a multiple of 6

13 a)

$$n^3 - n = n(n^2 - 1) = n(n-1)(n+1)$$

So it is three consecutive numbers positive integers: $n-1, n, n+1$

One of these must be a multiple of 3

So $n^3 - n$ must be a multiple of 3

13 b)

$$n^3 - n = n(n^2 - 1) = n(n-1)(n+1)$$

When $n = 3m$

$$(3m)^3 - (3m) = 3m(3m-1)(3m+1) \text{ which is divisible by 3}$$

When $n = 3m + 1$

$$\begin{aligned}&(3m+1)^3 - (3m+1)\\ &= (3m+1)(3m+1-1)(3m+1+1)\\ &= (3m+1)3m(3m+2) \text{ which is divisible by 3}\end{aligned}$$

When $n = 3m + 2$

$$\begin{aligned}&(3m+2)^3 - (3m+2)\\ &= (3m+2)(3m+2-1)(3m+2+1)\\ &= (3m+2)(3m+1)(3m+3) \text{which is divisible by 3}\end{aligned}$$

So $n^3 - n$ must be a multiple of 3

14)

Consider n is even: $n = 2k$

$$n^4 - 4 = (2k)^4 - 4 = 16k^4 - 4$$

This is 4 less than a multiple of 8, hence not divisible by 8

Consider n is odd: $n = 2k + 1$

$$\begin{aligned}n^4 - 4 &= (2k+1)^4 - 4\\ &= 16k^4 + 32k^3 + 24k^2 + 8k + 1 - 4\\ &= 8(2k^4 + 4k^3 + 3k^2 + k) - 3\end{aligned}$$

This is 3 less than a multiple of 8, hence not divisible by 8

$\therefore n^4 - 4$ is not divisible by 8